# SENTHIL KUMAR, AI & CLOUD ARCHITECT

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- 🧢 🐢 The hats I wear as an Applied AI Engineer | AI & Cloud Architect:
  - An AI Architect: Design agentic systems, RAG solutions, and reusable GenAI platforms
  - A Cloud Architect: Define and architect scalable AWS-based AI systems using serverless patterns
  - An Applied AI Engineer: Deliver NLP and deep learning solutions from concept to production

- ✍️ Passionate learner and technical writer: 40+ posts at senthilkumarm1901.github.io and 10+ posts on Medium
  - Example blog posts: ❖(1)LangGraph, Strands, and Hermes Comparison; ❖(2)Single LLMs to Agentic AI: GenAI Evolution Explained; ❖(3)Git: TimeTravel Masterclass; ❖(4)Leveraging LLMs in the Cloud

## WORK EXPERIENCE

**Lead ML Engineer, Toyota Connected India** **Jul 2022 – Present**

- Architect GenAI platforms, RAG systems, and agentic AI applications for enterprise use cases
- Lead cross-functional teams comprising BE, FE & AI engineers to deliver production-grade AI solutions
- Published 4 articles in TCIN Medium channel. Delivered a Talk on Serverless Python App in PyCon2023

**Lead Data Scientist, Ford, Analytics Team** **May 2018 – Jun 2022**

- Designed and developed NLP inference solutions in collaboration with global teams
- Built reusable, containerized AI components for production NLP applications

**Assistant Manager, LatentView Analytics (LV)** **Apr 2014 – Apr 2018**

- Roles:**NLP Data Scientist** + **Project Delivery Manager**; led a team of 8+ members

**Lead Analyst, Beroe Inc** **Jul 2010 – Dec 2013**

- Produced Market Research reports on how to procure indirect spend categories

## PERFORMANCE AWARDS

- **Toyota** – Employee of the Month (Sep‘23); 2nd place, AWS India Manufacturing Hackathon (‘24); Team Winner, DriveDesign (‘25 & ‘26), AWS Game Day Security Event (‘26), Hackcelerate (‘23 & ‘26)
- **Ford** – Asia-Pacific Recognition Award (May 2019)
- **LatentView** – Encore Award (Jul–Sep 2016 Quarter)
- **Beroe** – Knowledge Contributor Award (twice, 2013)

## EDUCATION

- Bachelor’s in Engg - Electronics - 8.6 CGPA | Madras Institute of Technology, 2006 - 2010
- State Topper in 2006 TN Engineering Entrance Exam (Physical Science-50/50); 12th Grade-95%, 10th Grade-92%

## TECHNICAL UPSKILLING

- Architecting Solutions on AWS, Coursera-AWS, Sep‘26
- A 3-Course Specialization on AWS GenAI & Bedrock, Coursera-AWS, Jan‘26
- Modern AI Agents, O’Reilly, Aug‘25 | Building Apps with Anthropic’s MCP, DeepLearning.AI, Jun‘25
- Git Skillcred, Linux Foundation, Jan‘25 | DevOps on AWS(4 courses), Coursera-AWS, Oct‘24
- Generative AI with LLMs, Coursera-Deeplearning.ai, Nov‘23
- DeepLearning Specialization(5 courses), Coursera-Deeplearning.ai, Dec‘18 - May‘19

**Skills**

- **Architecture**: AI Platform Design, Solution Architecture, Agentic AI, RAG Systems, Event-Driven Systems
- **Cloud**: AWS Security, Compute, Networking, Serverless (Lambda, API Gateway, DynamoDB), AWS CLI/SAM
- **Languages**: Python, SQL (basics), Markdown, Bash/Shell Scripting
- **AI Frameworks**: LangGraph, Strands, FastAPI, PyTorch, Transformers, SpaCy
- **Engineering**: Docker, Git, Python Environment Management, API Design
- **Leadership**: Technical Mentoring, Cross-Functional Collaboration, Solution Delivery

- 👀 **Curious what I have been cooking lately?** 🔗 LinkedIn Featured Section

## KEY PROJECTS

**Agentic AI Platform | Enterprise Productivity | 2026 | Ongoing**

```
User ↔ React–Tauri–Rust App: [Agent Loop ↔ LLM API | Tools | DB]
```

- **Impact**: Architecting a desktop-first Agentic AI Platform that enables non-developers to create, deploy, and monitor AI agents safely
- **Architecture**: React UI → Tauri IPC → Rust BE
  - Agent Loop inspired by hermes agent loop; architecture similar to Andrew Ng's OpenWorker
- **Tech Stack**: Rust | Tauri 2 | React + TypeScript + Vite | SQLite | Bedrock · Azure OpenAI · Ollama

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**GenAI Workflow Platform | Enterprise Productivity | 2024**

```
Foundation to GenAI apps: YAML Spec → WorkFlow Engine → LLM
Chaining/WorkFlow → Result
```

- **Impact**: Architected a Serverless GenAI Platform enabling multiple AI applications to be built on a common workflow framework
- **Architecture**: Defined a reusable LLM-workflow engine driven by declarative YAML specifications
- **Highlight**: Designed a provider-agnostic architecture supporting AWS Bedrock and Azure OpenAI through a unified gateway
- **Tech Stack**: Python | YAML | AWS (Lambda · S3 · Bedrock) | LLM Gateway (Bedrock + Azure OpenAI)

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**Restricted-Output Conversational AI Agent | Customer Service Automation | 2025–26**

```
User query → (FSM L1) "Decides WHAT" → (L2) "Does HOW" → LLM & Tools →
Templated Responses
```

- **Impact**: Built an explainable, template-controlled customer-service chatbot validated across three concurrent use cases (service booking, vehicle recall check, new car sales) on one orchestration backbone.
- **Architecture**: Designed a two-layer Finite State Machine system (FSM)
- **Tech Stack**: Client: React | API: Node.js | DB: MongoDB | BE: Python & YAML-declared FSM | LLM: OpenAI

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**PolicyBot: Agentic RAG Chatbot for Enterprise | HR · IT · InfoSec | 2025**

```
Employee Query → RAG Agent → Tool Call (Vector DB OR MCP Tool) → RAG
Agent Thinks:Confident? –Yes→ Compose Source–cited Answer | –No→
Rephrased Query to RAG Agent
```

- **Impact**: Delivered a role-aware, multilingual policy assistant with *every chunk source-cited* for traceability
- **Architecture**: Architected an enterprise-grade agentic RAG system on AWS
- **Technology Decisions**: Evaluated and adopted vector search, orchestration loop, and agent integration approaches balancing control and operational cost
- **Highlight**: Moved from Bedrock managed agent orchestration (`agent–runtime`) to self-orchestrated (`bedrock agentcore`) RAG for finer control over planning, prompts, and cost.
- **Tech Stack**: AWS Bedrock | AWS OpenSearch | MCP | OpenAPI Schema | Amazon RDS

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**Call Center Analytics Dashboard | Contact Center Intelligence | 2024**

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audio → (1) diarized turn–level transcript → (2) Prompt–configurable
metric extraction → persisted insights → dashboard
```

- **Impact**: Architected a speech-to-insight pipeline turning raw contact-center audio into agent-level performance metrics on a dashboard
- **Highlight**: Migrated from open-source PoC for diarization + STT and metric extraction to serverless managed APIs after cost-usage analysis
- **Tech Stack**: AWS Transcribe (diarization + STT) | LLM API (metric extraction) | AWS (Lambda · S3) | React UI

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